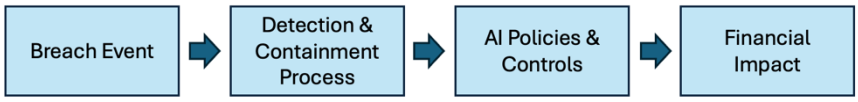
AI Security as a Cost-Saving Strategy for Companies

Elijah DuMond

I. Introduction

Each day over $24 billion dollars are lost by corporations globally due to fallout from a cyber-attack, that's nearly $17 million each minute. Whether directly stolen or just in losses of what could have been made without the shadow of the attack, this massive amount only seems to be on the rise due to the prevalence of AI in the cyber scene. The introduction to generative AI has lead to massive increases in cyber-attacks by smaller threat-actors instead of just the large known cyber-crime organizations. These attacks are becoming more frequent as well as more intricate and targeted. There seems to be many different factors that can influence the cost of a data-breach and finding out which factors could save a company the most money seems to be a question that many people are having. Some of these following factors focused in this paper are how a company using AI-Monitoring is effected versus a company that has human lead monitoring, as well as AI policies and controls set in place. Many different companies have yet to adapt to the artificial intelligence scene and have not set limits on how their employees are able to use these agents, and they could be increasing the risk of an attack due to the injection of confidential information. Another reason to dive into this is the rising cost of cyber-attacks each year in the United States, just one year ago in 2024 the average cost of a singular data breach globally was $4.88 million (IBM), while in 2025 it is said to be around $4.44 million. (IBM)

Despite this decrease globally, the United States actual saw an increase in this cost. In 2024 the average cost of an attack was $9.38 million (IBM), and it 2025 rose to $10.22 million (IBM). So is the best way to keep up with these rising numbers to fight back with AI? This project investigates how the use of AI in cyber-defense affects the time to detect and contain breaches, as well as overall financial impact. Using a simulated dataset based on IBM 2025 Cost of a Data Breach statistics, we explore relationships between AI deployment, breach characteristics, and cost outcomes. The dataset contains 10,000 simulated breach incidents that could be located in every region across the globe as well as containing key information on whether these companies had these different policies in place, how long it took to contain the breach, and whether AI was present on either side of the attack. Using visualizations, applied probability, statistical inference, and regression modeling, we are able to quantify the effect of these variables on the outcome of each attack.



(**figure 1**: simple outlook on how BI solution will be discovered and implemented)

II. Business Intelligence Methods

The BI framework described in this section details the roadmap, data architecture, and statistical methodologies that aimed to guide the industry toward a more insight driven, effective Business solution.

a. The Business Process

The first step in the BI effort is establishing a clear roadmap that outlines the plan to provide the industry with a better practice. The BI roadmap guiding this project focuses on transforming raw data from cyber breaches to actionable intelligence using different data management strategies. The roadmap supports execution and coordination within the project. Data would be collected through academic sources to be used in the process and data collection phase. The system would then be evaluated using linear and logarithmic regression to see whether the findings would be statistically significant and improve the current business solution in the cyber-security space.

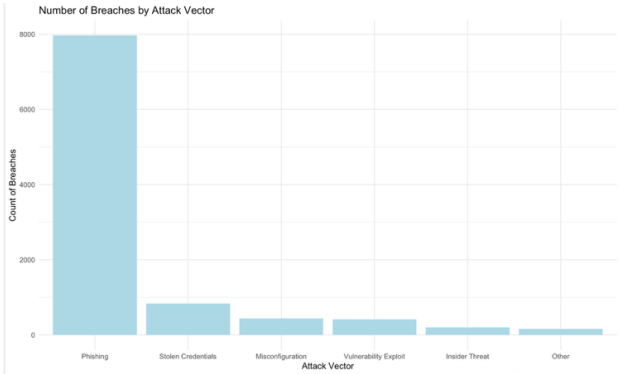
1. Key Deliverables	**2. Receive Sponsor**	**3. Timeframe**	**4. Process and Data Collection**
Improve company security against cyberattacks Save millions of dollars by implementing stronger guard rails and training employees thoroughly	Talk to IBM cloud security agent to learn more about his standing and get his support, and to better understand the IBM cost of a data breach report 2025	I will complete the research and models by December and present it Security officers and others who may interact with confidential data	I will collect data emphasizing the success of AI defense on saving companies money during cyber attacks
5. Data Subjects	**6. Technology Required**		
Looking at the yearly recap from IBM, I will simulate a data set to analyze and predict what could happen during cyber attacks	Rstudio, Excel, Python, along with academic journals and research articles		

(**figure 2:** BI roadmap developed before project was started)

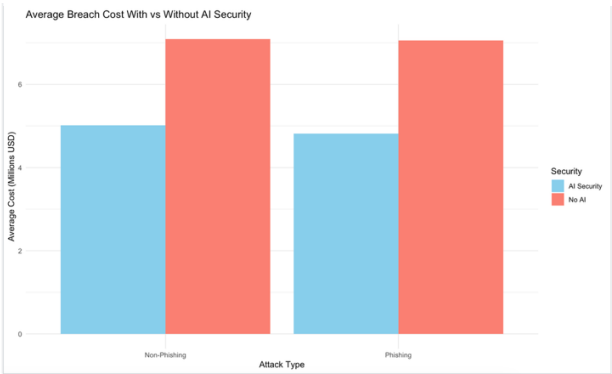
The study will assess the data obtained, by simulating a dataset using python to then use in RStudio. The data frame will closely resemble that of which companies like IBM and CISA could collect on a large scale of cyber-attacks, but their data is confidential and is unable to be obtained for a project such as this. The roadmap provided the sequential steps required to complete the key deliverables within the defined timeframe in the above figure.

b. Data and Data Structures

The data used in the study were mainly obtained from the IBM Cost of a Data breach report for 2025. The actual data is confidential and was not able to be obtained even for academic research opportunities. To work around the data availability constraint, Python was used to generate a synthetic dataset that could replicate the structure of which IBM could have the data collected and stored on each of their cyber-incidents on file (see appendix A). Since the data was crafted using python, the variables could be cleanly created and there will not need to be as much data management as necessary with publicly obtained sets. So common data problems such as missing values or incorrect values could be avoided. The main data management strategy used to clean the dataset used was changing character values of 'True' and 'False' to a Boolean value of TRUE or FALSE. This way, the variables are able to be used mathematically in different equations to create graphics and find average values using r scripts. Before moving forward with the project, it was necessary to make sure that the average values of each variable was within a reasonable amount of the average value stated by IBM in their report. Each value was near the true value granted some natural difference from the generations variance. In the generated dataset, each row was a cyber attack that would contain the following values: industry, region, company size, attack vector, attacker using AI, AI defense, AI policies at company, days to identify, days to contain, total days, extensive AI security, AI access controls, AI governance and finally cost of the breach.



(**figure 3**: count of cyber-attacks by attack vector)

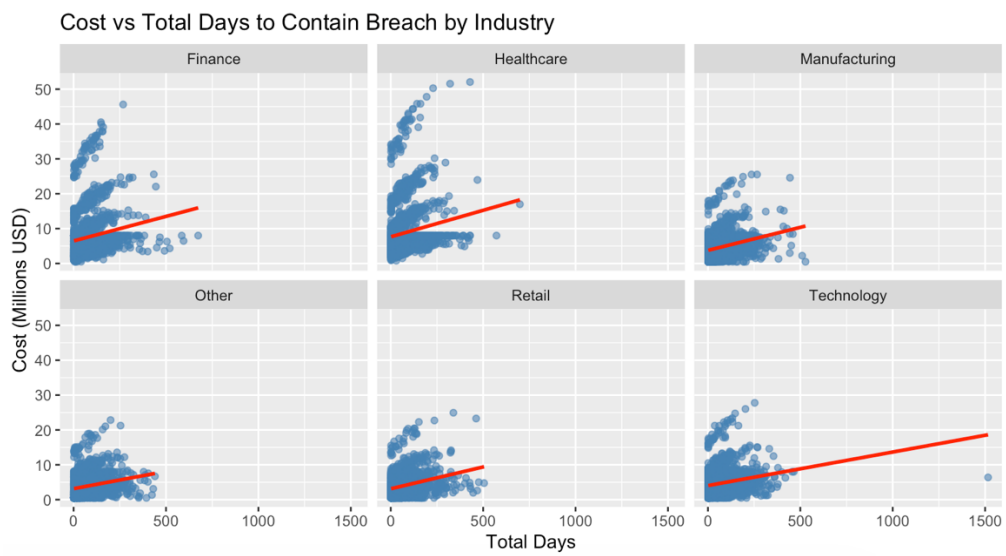


(**figure 4:** cost of non-phishing vs. phishing attacks)

Since the main goal of the project in the roadmap is to save companies money, finding major contributors to the cost was very important. One interesting finding during the data management section was what percentage of attacks were from the attack vector of phishing. Over 90% of all successful cyber-attacks are from phishing attacks. Because of this massive disparity it was interesting to figure out whether these attacks were more devastating than other vectors, but the findings seemed to show that the vector didn't have too much to do with cost. The main effect will be discussed more during the results section. But as seen in the figure 5, the cost doesn't seem to be too different for non-phishing attack's vs. phishing attacks.

c. Statistical Methodology

After data management visualizations were crafted to find interesting relationships between the value of a variable and the final cost of a data breach. The main variables focused on during the visualization phase were whether AI security was present in the breach and how it effected the cost. See figure 4 above to see how the presence of AI security corresponded both in non-phishing attack as well as phishing attacks. Applied probability was used to craft hypotheses during the hypotheses testing phase. Two-tail t tests were used during the hypotheses testing section because there was not very many publicly obtainable solutions on whether a difference was going to be found when analyzing the cost of a data breach given AI's involvement in said attack because the introduction of artificial intelligence in cyber-security is relatively new. After crafting hypotheses and running t-tests we are able to see if our null hypotheses were able to be rejected or whether they could be moved forward into the linear and logarithmic regression testing phase. Both linear and logistic regression were used to analyze the data crafted because there seems to be some linear nature in the graph but they didn't seem to score very well in the adjusted R squared value during regression testing, the results section will further investigate the findings of that phase.



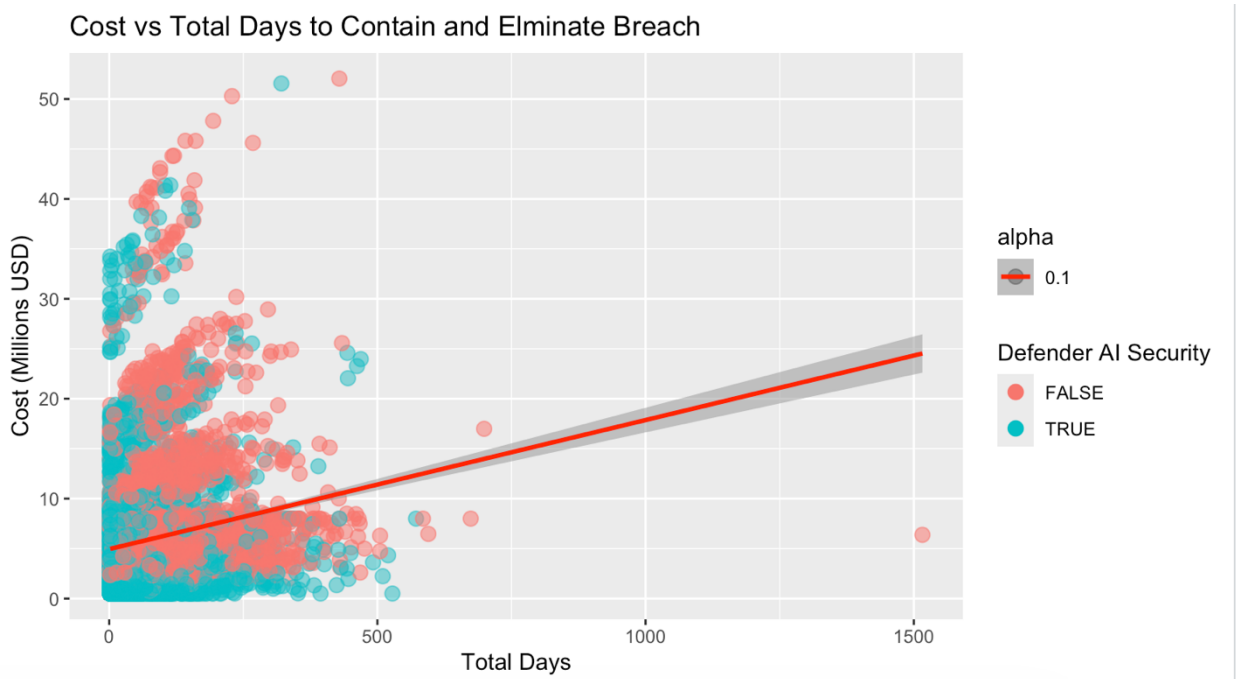
(**figure 5:** facet graph of each industry to see how the days seemed to effect the cost)

See figure 5 to understand how the data didn't score well linearly because a single line couldn't define the set well. Discussed later in the results section it was tricky to figure out which variables were causing this sectioning in each graph. So logarithmic regression was used to analyze the graphs because of the slight curve to each of the groupings on the graphs. The variables selected to undergo this testing were decided from the former probability, and hypothesis testing. Combining all of these data methodology the data will be manipulated to produce meaningful intelligence in the cyber-security industry, and will help shape the future years to come.

III. Results

The results presented in this section translate the statistical analyses and data exploration and management discussed in previous section into clear actionable findings that directly address the stated business needs in the crafted BI roadmap.

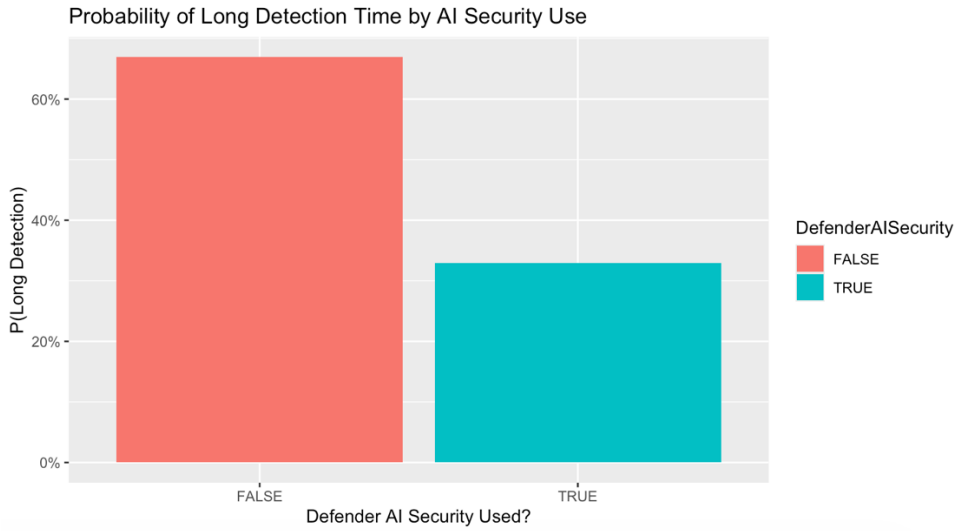
The first interesting thing that was focused on was finding out whether the attack vector was involved in contributing significantly to the cost of a cyber attack, this was quickly found to be not important, so artificial intelligence was brought into the conversation to be focused more on. Referencing back to figure 4 it is clear that AI security significantly lowers the cost of a cyber-attack regardless of the attack vector defined.



(**figure 6:** scatter plot of all cyber attacks total days, and the cost)

After creating basic visualizations there seemed to be a problem. As you can see in figure 6 there seems to be a few different clusters of points, and it was expected to be due to the different industries. During the simulation of the dataset it was defined that the most expensive industries were Finance and Healthcare, but as shown in figure 5, each industry continued to have issues with the clustering ruining the possibility of a linear regression to be successful. Through all of the data analysis no variable seemed to be the main culprit of causing that issue to occur, and it

may need to be researched more in the future to figure out what from the simulation was causing the weird clustering to occur because it was also disproven to be the region as well.



(**figure 7**: Graph showing the probability of long detection given AI security present)

Another interesting instance was how the length of a breach effected the cost of the breach in the end. Using applied probability it is observable that the probability that an attack's length was above the median length was 67% given that the company did not have any AI security present during the data breach (see figure 7).

Because the main goal was to find the meaningful factors that contributed to the cost of a cyber attack, but most of the points that AI improved was the amount of time. We were able to find out that the average cost of a breach when it was a long detection was $7.36 million while a quicker containment resulted in an average of $5.14 million. Long detection is defined as a timeframe that is greater than the median. After finding the probabilities of long detection hypotheses began to be crafted and tested. A goal of a (p value< .05) was used to define statistical significance. The main null hypothesis used for testing was "Use of AI monitoring does not have an impact on the probability of a long detection in a cyber-attack". A two sample t-test was used to reject the null hypothesis with a p-value of 2.2e-16 meaning that a defender using AI security plays a statistically significant part in determining how much time it takes to find and contain a cyber attack.

This hypothesis was the main hypothesis used to create the different regression models used during the regression model testing phase of the project. 9 different regressions were run on different variables looking at both the cost and time given many different AI presence variables. Both linear regressions and logistical regressions were used because the linear regressions didn't seem to be scoring very well due to the fact that there were different groupings of lines. During the search of what variable could be causing the cluster many different variables were tested but no avail. Industry, region, and company size all resulted in the same grouping showing up across

all facets produced. See appendix B to see the R file that explores the difficulty and trials trying to figure out which of the variables was causing the awkward grouping in the scatterplots, both the faceted and the accumulative. The grouping then had to be ignored during the regression testing and resulted in the poor scores for accuracy in R-squared and adjusted R-squared.

Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
4.9487563	7.7458869	4.5875761	4.781898	7.04441	14.68006	6.078381	13.0256	2.17606
2.00E-16	2.00E-16	2.00E-16	2.00E-16	2.00E-16	8.09E-01	3.35E-01	2.00E-16	2.00E-16
n/a	2.00E-16	2.00E-16	2.00E-16	3.16E-01	2.00E-16	7.89E-01	5.78E-02	1.28E-01
n/a	2.00E-16	2.00E-16	n/a	0.819	2.00E-16	0.575	0.00513	0.0659
n/a	2.00E-16	n/a	n/a	n/a	2.00E-16	0.339	2.00E-16	2.00E-16
n/a	2.00E-16	n/a	n/a	n/a	2.00E-16	0.55	n/a	n/a
n/a	2.00E-16	n/a	n/a	n/a	2.00E-16	0.974	n/a	n/a
n/a	2.00E-16	n/a	n/a	n/a	2.00E-16	0.518	n/a	n/a
n/a	n/a	n/a	n/a	n/a	2.00E-16	0.332	n/a	n/a
n/a	n/a	n/a	n/a	n/a	2.00E-16	n/a	n/a	n/a
0.0348	0.1915	0.1826	0.0432	0.0433	0.3911	0.0003	0.4037	0.6672
0.0347	0.1909	0.1823	0.0430	0.0430	0.3905	-0.0005	0.4034	0.6671
360.70	338.10	744.20	225.40	150.60	641.70	0.41	1691.00	5010.00
Meh	not great	not good	not very good	same as last	pretty good	BADDDD	Decent	BEST

(**figure 9:** table of all regression along with the relevant p values and R^2 and adjusted R^2)

Each of the regressions was tracked in a table to evaluate the adjusted R^2 values of each test against the prior and following regression run, Model 9 was the logistic regression that was run and it scored the best adjusted R-squared value and that was a 0.6671. Although not great, it shows that 66.71% of all data could be explained by the predictors, and each of the predictors seemed to be statistically significant. The goal was to explain DaysToContain given: AI security, AI access controls, AI governance, and Extensive AI used. AI security is defined as some sort of AI monitoring used to analyze the logs of the company and pick out when something seems wrong. AI access controls is whether the company breached has any controls set in that there company is unable to access certain parts of AI, while governance is whether they are even allowed to use AI at their work or for example can't copy and paste customer or account information into the agent. Finally extensive use, would be more than just the AI monitoring and would also contain active moderation and interaction from AI. Despite the awkward clustering of the graph ,about 67% of all of the Days to contain points could be explained by the given variables regarding AI's presence during a cyber-attack.

Taken together, these results discussed provide a view of the relationships among the key variables regarding the presence of AI during a cyber-attack and the cost given that containment time was the leading factor in the cost of a breach. These findings indicate a few patterns that warrant some further examination. The main finding in our research was that AI security can save your company on average about $1.1 million dollars if involved in a cyber-attack, while basic AI monitoring tools could cost as low as $1000 dollars per month but stronger solutions such as IBM are about $54,000 per year. A yearly cost of $54,000 to potentially save millions in a breach. Although not 100% accurate, there seems to be some correlation, and confirm the key deliverable of saving companies money.

IV. Conclusions

The analyses conducted in this Business Intelligence project provide an encompassing view of how artificial intelligence can affect the outcomes of cyber-attacks and the associated financial costs, due to the detection and containment times correlated. Using a simulated dataset trained on the IBM Cost of a Data Breach report, this study would evaluate 10,000 cyber-attack incidents across many industries, regions and different attack vectors, with the key objective to understand how AI involvement would correlate. This also included looking at associated policies and governance regarding artificial intelligence and how all of this ultimately impacted the cost incurred by a targeted organization during a breach. The results consistently indicate that AI security has a significant effect on both the time to contain breaches as well as the cost of those breaches. This may need to be further improved upon using a different type of significance test because now there is some baseline results and evidence that it may save money and a single tail test could be performed.

A central finding from the data analysis is that the presence of AI in monitoring and security significantly reduce breach detection and containment time. Applied probability and hypothesis testing revealed that companies without AI security were 67% more likely to experience a long detection period, defined as exceeding the median breach containment time, while companies using AI monitoring experienced a substantially quicker containment. This reduced detection and containment time translates directly into financial savings, with an average reduction of $1.1 million per breach when AI security is implemented. Given the relatively low cost of AI monitoring solutions, these findings demonstrate a high return on investment for organizations seeking to minimize the financial losses during a potential cyber-attack. Another finding that may be interesting in the future is how less likely a breach is to occur if a company is using AI monitoring and security because the data-sets that are released are about successful cyber-attacks only. Finding a dataset that contains failed breaches as well would greatly improve the case for AI security if a similar finding happened to appear there as well.

Regression analyses further confirmed the significance of AI-related variables. Linear and logistic regression were applied to evaluate predictors such as AI security presence, AI access controls, AI governance, and extensive AI usage. Despite challenges with the clustered data appearing in the scatter plot, and the imperfect model fit, the logistic regression model explained approximately 67% of the variance in Days to Contain, with all predictors in the final logistic regression demonstrating statistical significance. These results indicate that AI interventions, when combined with some governance and access controls within a company, not only are they effective, but they can be used as a way to improve their resilience and posture in the cyber scene. The results also show that factors initially assumed to drive cost such as attack vector and industry, were less impactful than the presence and management of AI security. This differences highlights the emerging importance of AI defensive risk strategies in the modern cyber landscape.

While the findings were interesting, several limitations should be outlined. The dataset that was simulated to replicate the IBM Cost of a Data Breach report, because that dataset was confidential. This simulation allowed for full control over data quality and variable decisions. But as a result of this it may not have captured all of the nuance present in real life incident, such as the organizations behavior, or the attacker sophistication or skill. In addition, data clustering limited the linear regression model's predictive accuracy, suggesting that additional research could be needed to figure out the variables that may affect containment time and breach costs. The analysis also assumed consistent definitions of AI security and governance, where in real life companies could have widely different governance rules or AI security agents, or platforms. Despite these constraints, the statistical methods selected, ranging from applied probability to regression modeling, were appropriate for testing hypotheses created and they can provide insight into the relationships that AI might have on breaches in the real world outside of the simulation.

From a business perspective, the findings in the BI project have immediate and long-term implications. Organizations that adopt AI monitoring tools, and begin to enforce stricter access controls and governance for AI, not only reduce the probability that a cyber-attack goes undetected for long amounts of time, but could potentially save over a million dollars for each cyber-attack that they are involved in. As well as, this project provides a framework for continuous improvement, as companies can use their own BI solutions to monitor the effectiveness of AI policies within their own practice. By combining simulated analytics with actionable BI insight, this study illustrates how organizations can prioritize AI security investments to mitigate risk, enhance efficiency, and align cybersecurity strategies with their financial objectives.

As we move into the future, the cyber defense eco system will continue to evolve alongside the developments in artificial intelligence. As attackers continue to leverage AI for their own gain conducting attacks, companies will need to integrate AI defensive strategies to gain a competitive advantage both in security as well as their cost management. Ultimately investing in AI-driven cybersecurity could potentially reduce financial losses, strengthen organizational cyber-resilience as well as improve companies response capabilities.

V. Appendices

Appendix A: Python Data Preparation Script

This script uses the average values from the IBM Cost of a Data Breach 2025 Report and simulates 10,000 cyber-attacks with the values corresponding to the different possible outcomes.

```
import numpy as np
import pandas as pd
import random
# ------------------------
# Config with IBM 2025 values
# ------------------------
RANDOM_SEED = 42
np.random.seed(RANDOM_SEED)
random.seed(RANDOM_SEED)

n = 10000  # number of simulated breaches

industries = ["Healthcare", "Finance", "Retail", "Technology", "Manufacturing",
"Other"]
industry_probs = [0.2, 0.2, 0.15, 0.15, 0.15, 0.15]

regions = ["USA", "Europe", "Asia-Pacific", "Latin America", "Middle East & Africa"]
region_probs = [0.25, 0.25, 0.2, 0.15, 0.15]

company_sizes = ["Small", "Medium", "Large"]
company_size_probs = [0.55, 0.35, 0.10]

attack_vectors = ["Phishing", "Stolen Credentials", "Misconfiguration",
                  "Vulnerability Exploit", "Insider Threat", "Other"]
attack_weights = [0.9, 0.1, 0.05, 0.05, 0.03, 0.02]  # Phishing dominant

industry_cost_multiplier = {
    "Healthcare": 1.9,
    "Finance": 1.6,
    "Retail": 0.9,
    "Technology": 1.0,
    "Manufacturing": 1.0,
    "Other": 0.85
}

global_avg_cost = 4.44
usa_avg_cost = 10.22

attacker_ai_prob = 0.16

rows = []
for i in range(n):
    company_id = f"CMP{i+1:05d}"
    industry = np.random.choice(industries, p=industry_probs)
    region = np.random.choice(regions, p=region_probs)
    size = np.random.choice(company_sizes, p=company_size_probs)
```

```
    attack_vector = np.random.choice(
        attack_vectors,
        p=np.array(attack_weights) / sum(attack_weights)
    )

    ransom_paid = attack_vector == "Ransomware/Malware" and np.random.rand() < 0.35

    ai_security = np.random.rand() < 0.5
    ai_governance = np.random.rand() > 0.63
    ai_access_controls = np.random.rand() > 0.97
    attacker_ai = np.random.rand() < attacker_ai_prob

    # Detection and containment times
    days_to_identify = int(np.random.lognormal(mean=4.5, sigma=0.6))
    days_to_contain = int(np.random.lognormal(mean=2.5, sigma=0.6))

    # AI security reduces time by ~80 days total (IBM 2025)
    if ai_security:
        reduction = 80
        days_to_identify = max(1, days_to_identify - int(reduction * 0.6))  # ~48
days
        days_to_contain = max(1, days_to_contain - int(reduction * 0.4))    # ~32
days

    # Additional reduction for extensive AI use
    ai_extensive_use = np.random.rand() < 0.3
    if ai_extensive_use:
        days_to_identify = max(1, days_to_identify - 50)
        days_to_contain = max(1, days_to_contain - 30)

    total_days = days_to_identify + days_to_contain

    base_cost = usa_avg_cost if region == "USA" else global_avg_cost
    size_multiplier = {"Small": 0.6, "Medium": 1.0, "Large": 1.8}[size]
    time_penalty = min(1 + (total_days / 500), 1.5)

    mean_cost_m = base_cost * industry_cost_multiplier[industry] * size_multiplier *
time_penalty

    if ai_security:
        mean_cost_m -= 1.9  # IBM-reported savings

    if ai_extensive_use:
        mean_cost_m *= 0.9

    mean_cost_m = max(0.5, mean_cost_m)
    total_cost_m = max(0.5, np.random.normal(loc=mean_cost_m, scale=0.8))

    # Cap global breach costs to reduce average
    if region != "USA":
        total_cost_m = min(total_cost_m, 8.0)

    rows.append({
        "CompanyID": company_id,
```

```
        "Industry": industry,
        "Region": region,
        "CompanySize": size,
        "AttackVector": attack_vector,
        "RansomPaid": ransom_paid,
        "AttackerAIUsed": attacker_ai,
        "DefenderAISecurity": ai_security,
        "AIControlsMissing": ai_access_controls,
        "AIPolicyLacking": ai_governance,
        "DaysToIdentify": days_to_identify,
        "DaysToContain": days_to_contain,
        "TotalDays": total_days,
        "AI_Extensive_Use": ai_extensive_use,
        "CostMillionsUSD": round(total_cost_m, 2)
    })

df_attacks = pd.DataFrame(rows)
df_attacks.to_csv("simData.csv", index=False)
print("Dataset saved as simData.csv")
```

Appendix B: R Data Analysis Script

This script contains the data management required to turn the data frame into information and effective visualizations. The code for the visualizations is also present in the script.

```
##################################################
# Title: The Use of AI in Cyber-defense
# Author: Elijah DuMond
# Description: Prove that the use of AI in cyber-defense can help save
# companies millions and can lead to faster detection and less victims.
##################################################

rm(list=ls())
library(tidyverse)
library(readxl)
library(ggplot2)

#IMPORTING the simulated data set that I used
#The IBM cost of a databreach report to get the values
#Since I simulated the data set, much less data management
# is required because I got to name each column and curate
# the generated values in each one.
df <- read.csv("~/Downloads/simData.csv")

#characterized variables from the python file that I created
#############################################################
# Dataset: simData.csv
# Description: Simulated cyber breach data based on IBM 2025
# values to explore AI use in cyber-defense and its impact
# on breach detection, containment, and costs.
#############################################################

# Variables:

# CompanyID: Unique identifier for each company (string)

# Industry: Industry category of the company (factor)
#   Levels: Healthcare, Finance, Retail, Technology, Manufacturing, Other

# Region: Geographic region of the company (factor)
#   Levels: USA, Europe, Asia-Pacific, Latin America, Middle East & Africa

# CompanySize: Size of the company (factor)
#   Levels: Small, Medium, Large
```

```
# AttackVector: Method used by attacker (factor)
#   Levels: Phishing, Stolen Credentials, Misconfiguration,
#           Vulnerability Exploit, Insider Threat, Other

# RansomPaid: Whether ransom was paid (logical, TRUE/FALSE) I end up removing this because it is irrelevant in my
research

# AttackerAIUsed: Whether the attacker used AI driven attacks (logical, TRUE/FALSE)

# DefenderAISecurity: Whether the company used AI monitoring for defense (logical, TRUE/FALSE)

# AIControlsMissing: Whether AI access controls were missing for employees(logical, TRUE/FALSE)

# AIPolicyLacking: Whether AI governance/policy was lacking at the company (logical, TRUE/FALSE)

# DaysToIdentify: Number of days to identify the breach (integer)

# DaysToContain: Number of days to contain the breach (integer)

# TotalDays: Total days to identify + contain the breach (integer)

# AI_Extensive_Use: Whether the company used AI extensively, similar to monitoring, but more advanced
systems(logical, TRUE/FALSE)

# CostMillionsUSD: Total cost of the breach in millions USD (numeric)
##############################################################


#Convert True/False strings to logical TRUE/FALSE
df$DefenderAISecurity <- df$DefenderAISecurity == "True"
df$AI_Governance <- df$AIPolicyLacking == "False"
df$AIAccessControls <- df$AIControlsMissing == "False"
df$AttackerAIUsed <- df$AttackerAIUsed == "True"
df$AI_Extensive_Use <- df$AI_Extensive_Use == "True"
df$RansomPaid <- NULL

# Making sure simulated dataset is similar to IBM findings in 2025
# Overall average cost (should be near 4.44M)
mean(df$CostMillionsUSD)
# Average cost in USA (should be near 10.22M)
mean(df$CostMillionsUSD[df$Region == "USA"])
# Average cost outside USA (should be closer to global ~4.44M)
mean(df$CostMillionsUSD[df$Region != "USA"])
# Average cost with AI security (should be ~1.9M lower than without)
mean(df$CostMillionsUSD[df$DefenderAISecurity == TRUE])
mean(df$CostMillionsUSD[df$DefenderAISecurity == FALSE])
# Percentage of breaches where attackers used AI (should be ~16%)
mean(df$AttackerAIUsed) * 100
# Percentage of companies with AI governance policies (~37%)
mean(df$AI_Governance) * 100
# Percentage of companies with AI access controls (~3%)
mean(df$AIAccessControls) * 100

# DATA VISUALIZATIONS

##Prototyping visuals
#1 Define Practice Changing Metrics
#2 List Variables
#3 Pick an outcome
#4 draw your visual
#5 Identify focal point

#create bar chart of attack vector counts
ggplot(df, aes(x = reorder(AttackVector, -table(AttackVector)[AttackVector]))) +
  geom_bar(fill = "lightblue") +
  labs(
    title = "Number of Breaches by Attack Vector",
    x = "Attack Vector",
    y = "Count of Breaches"
  ) +
  theme_minimal()
```

```
#filter the phishing and non-phishing
phishing_ai <- df %>% filter(AttackVector == "Phishing", DefenderAISecurity == TRUE)
phishing_no_ai <- df %>% filter(AttackVector == "Phishing", DefenderAISecurity == FALSE)

non_phishing_ai <- df %>% filter(AttackVector != "Phishing", DefenderAISecurity == TRUE)
non_phishing_no_ai <- df %>% filter(AttackVector != "Phishing", DefenderAISecurity == FALSE)

#create the data frame
costs <- data.frame(
  category = c("Phishing", "Phishing", "Non-Phishing", "Non-Phishing"),
  ai = c("No AI", "AI Security", "No AI", "AI Security"),
  avgcost = c(
    mean(phishing_no_ai$CostMillionsUSD, na.rm = TRUE),
    mean(phishing_ai$CostMillionsUSD, na.rm = TRUE),
    mean(non_phishing_no_ai$CostMillionsUSD, na.rm = TRUE),
    mean(non_phishing_ai$CostMillionsUSD, na.rm = TRUE)
  )
)
ggplot(costs, aes(x = category, y = avgcost, fill = ai)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Average Breach Cost With vs Without AI Security",
    x = "Attack Type",
    y = "Average Cost (Millions USD)",
    fill = "Security"
  ) +
  scale_fill_manual(values = c("No AI" = "salmon", "AI Security" = "skyblue")) +
  theme_minimal()

#Visualization to see how cost distribution is effected by ai defender security
ggplot(df, aes(x = DefenderAISecurity, y = CostMillionsUSD, fill = DefenderAISecurity)) +
  geom_boxplot() +
  labs(title = "Cost Distribution by AI Security Use", y = "Cost (Millions USD)")

#Scatter plot to show positive relationship between time and cost
ggplot(df, aes(x = TotalDays, y = CostMillionsUSD, color = DefenderAISecurity, alpha = .1)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", color = "red") +
  labs(
    title = "Cost vs Total Days to Contain and Elminate Breach",
    x = "Total Days",
    y = "Cost (Millions USD)",
    color = "Defender AI Security"
  )
#It looks like there is some clustering for each of the industries
unique(df$Industry)
#There are 6 industries but 4 clusters
#Faceted Scatterplots by industry
ggplot(df, aes(x = TotalDays, y = CostMillionsUSD)) +
  geom_point(alpha = 0.6, color = "steelblue") +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  facet_wrap(~ Industry) +
  labs(
    title = "Cost vs Total Days to Contain Breach by Industry",
    x = "Total Days",
    y = "Cost (Millions USD)"
  )
#Healthcare and finance seem to be similar graphs and the
#rest seem to be similar so I am making 2 groups
df$group <- ifelse(df$Industry %in% c("Finance", "Healthcare"), 1, 0)

#No significant difference in the industries
#Grouping Finance and Healthcare for a regression line



#APPLIED PROBABILITY
#Within simulated dataset did AI security use result in faster fixes
df <- df %>%
  mutate(LongDetection = DaysToIdentify > median(DaysToIdentify, na.rm = TRUE))

prob_long <- df %>%
  group_by(DefenderAISecurity) %>%
```

```
  summarize(
    P_LongDetection = mean(LongDetection, na.rm = TRUE)
)
prob_long

#long detection is anything over the median
ggplot(prob_long, aes(x = DefenderAISecurity, y = P_LongDetection, fill = DefenderAISecurity)) +
  geom_col() +
  labs(
    title = "Probability of Long Detection Time by AI Security Use",
    x = "Defender AI Security Used?",
    y = "P(Long Detection)"
  ) +
  scale_y_continuous(labels = scales::percent)

#Average cost of attack if AIDefense is or isn't present
df %>%
  group_by(DefenderAISecurity) %>%
  summarise(
    AvgCost = mean(CostMillionsUSD, na.rm = TRUE),
    Count = n()
  )

#probability that a successful attack is phishing
phish_overall <- df %>%
  summarize(
    total_attacks = n(),
    phishing_attacks = sum(AttackVector == "Phishing", na.rm = TRUE),
    P_phishing = phishing_attacks / total_attacks
  )
phish_overall
#according to CISA about 90% of succesful attacks are phishing

#We are going to look at if Longer DaysToContain results in a average higher cost
df <- df %>%
  mutate(LongContainment = DaysToContain > median(DaysToContain, na.rm = TRUE))

cost_summary <- df %>%
  group_by(LongContainment) %>%
  summarize(
    AvgCost = mean(CostMillionsUSD, na.rm = TRUE),
    Count = n()
  )

cost_summary

#HYPOTHESIS TESTING and STATISTICAL INFERENCE

#Null Hypothesis = Use of AI monitoring does not have an impact on the average cost of a data breach
#Alternative Hypothesis = Use of AI monitoring does have an impact on the average cost of a data breach

#C = cost
#AC = average cost
#AI = 1 if present, 0 if not present

#H0 = C if(AI = 1) = AC
#HA = C if(AI = 0)  ≠ AC

#t tests
t_test_resultAICost <- t.test(CostMillionsUSD ~ DefenderAISecurity, data = df)
t_test_resultAICost

# Null Hypothesis = Use of AI monitoring does not have an impact on the probability of a long detection
# Alternative Hypothesis = Use of AI monitoring does have an impact on the probability of a long detection

# LD = LongDetection (1 if DaysToIdentify > median, 0 otherwise)
# P_LD = average probability of long detection
# AI = 1 if DefenderAISecurity is present, 0 if not present

# H0: P_LD | (AI = 1) = P_LD | (AI = 0)
# HA: P_LD | (AI = 1) ≠ P_LD | (AI = 0)

#t tests
```

```
t_test_resultAIDetection <- t.test(LongDetection ~ DefenderAISecurity, data = df)
t_test_resultAIDetection

#REGRESSION

#Logistic regression and Multilinear Regression

#Model of Cost in correlation to total days
model1 <- lm(CostMillionsUSD ~ TotalDays, data = df)
summary(model1)

#Model of Cost in correlation to days, ai defense used, industry
model2 <- lm(CostMillionsUSD ~ TotalDays + DefenderAISecurity +
               Industry, data = df)
summary(model2)

#Model of Cost in correlation to days, ai defense, and industrygroup
model3 <- lm(CostMillionsUSD ~ TotalDays + DefenderAISecurity + group, data = df)
summary(model3)

#Model of Cost in correlation to days to identify and days to contain
model4 <- lm(CostMillionsUSD ~ DaysToIdentify + DaysToContain, data = df)
summary(model4)

#Model of AI involvement, access controls, ai defense, and extensive ai security
model5 <- lm(CostMillionsUSD ~ DefenderAISecurity + AIControlsMissing + AIPolicyLacking, data = df)
summary(model5)

#Model to see if AI security has an effect on ai attacks
model6 <- lm(CostMillionsUSD ~ AttackerAIUsed * DefenderAISecurity + Industry + CompanySize, data = df)
summary(model6)

#Model to see if time is explained by attacker being AI and Industry
model7 <- lm(DaysToContain ~ AttackerAIUsed + Industry + CompanySize, data = df)
summary(model7)

#Model to see if time is explained by all AI settings
model8 <- lm(DaysToContain ~ DefenderAISecurity + AIAccessControls + AI_Governance + AI_Extensive_Use, data = df)
summary(model8)

#Model 8 seems to be the best at describing my data linearly despite my data not being
# very linear, there is maybe a better way to run a regression to find out what better fits my dataset

#Model 9 will be model 8 but as a logarithmic regression
model9 <- lm(log(DaysToContain) ~ DefenderAISecurity + AIAccessControls + AI_Governance + AI_Extensive_Use, data
= df)
summary(model9)

#The linear regression seems to explain my data best, but it still is not an amazing R^2


#I found during my research that days to contain is the largest variable on the cost of the
#most part so finding what could cause the daystoContain to increase was what I was focusing on
#during my regression testing.
#According to this from earlier, we can make the inference that DaysToContain influences Cost
#and we are able to see what could possibly influence the daysToContain
# LongContainment     AvgCost Count
#    <lgl>             <dbl> <int>
#  1 FALSE              5.14  6302
#  2 TRUE               7.36  3698
```